

Sriharsh Bhyravajjula

+1 (206) 751-4842 | Seattle WA

[sɪˈhɑːʃ bʰaɪrəˈvəʃːʊlə] | bhyrava@uw.edu | [linkedin.com/sbhyravajjula](https://www.linkedin.com/sbhyravajjula) | darthbhyrava.com

EDUCATION

University of Washington

Seattle, WA

MS in Computational Linguistics, 3.9/4.0

Sep. 2022 – Mar. 2024

Relevant Courses: Deep Learning, Machine Learning for NLP, Syntax Engineering, Shallow and Deep Techniques in NLP

International Institute of Information Technology

Hyderabad, India

B.Tech in Computer Science + MS by Research in Computational Linguistics, 3.2/4.0

Aug. 2014 – Aug. 2019

Relevant Courses: Data Structures, Algorithms, Statistical Methods in AI, Natural Language Processing, Speech Systems

- **Teaching Assistant** for CSE472: Natural Language Processing — Monsoon 2017, Monsoon 2018
- **Dean's Merit List Awardee** for exceptional academic performance — Monsoon 2017, Spring 2018

EXPERIENCE

Outreach Corporation

Seattle, Washington

Senior Applied Scientist

Oct. 2023 – current

- Building a custom **AI evaluation suite** by adding **LLM tracing and observability** stacks to Outreach's **AI agents and workflows** with use-case specific **metrics and telemetry design**, supporting both **human-in-the-loop** and **LLM-as-a-judge** with **evaluation-of-evaluators** approaches.
- **Won the Outreach 2024 Hackathon** in the Novel AI category for building a custom AI agent to assess current sales deal state, predict possible points of failure, and suggest mitigating actions.
- Improved Outreach's **LLM-based retrieval augmented generation (RAG)** pipeline using **temporally evolving multi-modal data sources** as context for **Kaia** Smart Deal Assist, Outreach's user-facing solution which allows users to ask nuanced questions about different aspects of an entire Outreach sales deal.
- Built an **LLM evaluation framework** to leverage human annotation for evaluating the quality of **Kaia** Smart Meeting Assist (SMA) - an **RAG** solution which allows users to ask questions about individual sales meetings.

Applied Science Intern

Jun. 2023 – Sep. 2023

- Using **in-context instruction learning** techniques in **prompt design** for **LLMs** to summarize answers to sales methodology questions from meeting transcripts.

Klevu AI

Hyderabad, India

NLP Data Scientist II

Jan. 2021 – Aug. 2022

- Built Klevu's ecommerce **knowledge graph** to semantically understand and answer complex search queries by creating APIs which curate information from domain-rich sources (product reviews, knowledge-bases, user analytics, and catalogs), and pool this data into custom **typeDB** databases capable of real-time inference.
- Analysed **product reviews** to enhance Klevu's ecommerce catalogs with product use-cases and feature sentiment, providing retailers with premium Business Intelligence (BI). Used a **BiLSTM model** in a grid-tagging scheme to label **aspect-opinion-sentiment triplets** in reviews with an **F1-score of 0.68** across two test domains.
- Reduced **zero-result searches** and improved **search conversion** by building and deploying an end-to-end synonym-enrichment API. Given product-names from store catalogs, the API extracts corresponding synonyms from the Open Multilingual Wordnet, and disambiguates their synset glosses against product descriptions with **Transformer-based embedding models** like **multilingual universal sentence encoder** and **mT5**.

NLP Data Scientist

Jul. 2019 - Dec. 2020

- Designed and built backend for Klevu's **Enhanced Search Analytics**, which featured in **Forbes** and helped ecommerce retailers understand shopper behavior and boost sales. This involved creation of (1) a scalable Elasticsearch stack for 1.5B+ data points (clicks, searches, checkouts), (2) NLP/ML scripts extracting BI like **ecommerce conversion metrics**, intuitive **word-graphs for search patterns**, and **time-series predictions** in real-time, and (3) interactive dashboards where these insights were served concurrently for 2000+ ecommerce retailers.

PUBLICATIONS

[so much depends / upon / a whitespace: Why Whitespace Matters for Poets and LLMs](#)

Sriharsh Bhyravajjula, Melanie Walsh, Anna Preus and Maria Antoniak — EMNLP (Main) 2025 : Suzhou, China

- Analyzed whitespace as a semantic-spatial element in poetry using 19k Poetry Foundation poems by 4k poets, releasing 2.8k public-domain texts with original formatting. Created WISP, a novel whitespace typology, and released WISP-Bench, a benchmark to evaluate whitespace performance in linearization methods that capture image-to-text, highlighting the impacts of dataset preparation on NLP and LLM tokenization design strategies.

[MARCUS: An Event-Centric NLP Pipeline that generates Character Arcs from Narratives](#)

Sriharsh Bhyravajjula, Ujwal Narayan and Manish Shrivastava — Text2Story @ ECIR 2022 : Stavanger, Norway

- Won the **Best Paper award** at the workshop and was given a **Student Travel Grant** by IIIT-H.
- Built a novel pipeline that (1) used a BiLSTM tagger to label **events** and a Semantic Role Labeller to identify **participant characters** from long-form narratives, (2) used fine-tuned BERT and RoBERTa models to extract proxy markers of emotion and sentiment to measure a **character's circumstance**, and (3) plotted smoothed **character arcs** as an aggregated change of character circumstance across event sequences.

PROJECTS

[Quantifying Event Saliency in Narratives](#)

Master's Thesis, University of Washington – Advisor: Dr. Maria Antoniak

- Posited saliency of events in narratives as a measure of event correspondence between original text and aligned sentences from abstractive summaries - where alignment was posed as a **similarity ranking problem** tackled using **SentenceTransformer** embeddings with a **Mean Reciprocal Rank of 0.81**. Also used **few-shot prompting** with **GPT4**, **Llama**, and **Claude** to improve results and explore event frame semantics.

[Prediction of Online News Article Popularity](#)

IIIT-H Course Project: Statistical Methods in Artificial Intelligence

- Predicted popularity of news articles for a local news agency by classifying potential impression count into threshold buckets. Reduced noisy feature space using **PCA**, and applied ML methods like **Naive Bayes**, **Logistic Regression**, **Random Forest**, **SVMs**, and **MLPs** with **accuracy of 53%-85%**.

TECHNICAL SKILLS

Programming & Scripting: python, C/C++, java, shell, L^AT_EX, javascript, HTML/CSS

DL/ML/NLP Frameworks: torch, sklearn, huggingface, stanza, spacy, nltk, tomatopy, gensim, *tensorflow*, *keras*

LLM/Agents: huggingface, llamaindex, langchain, langgraph, langsmith, openai, autogen, safetensors, GGUF

Data Management: pandas, elasticsearch, mongoDB, typedb, sql, neo4j

Deployment: gcp, aws, flask, django, torchserve, docker, ghost, jekyll, *tensorflow/serving*, *kubernetes*, *react*

IDE & Version Control: VS code, sublime, pycharm, jupyterlab, git, *gerrit*, *Eclipse*

POSITIONS OF RESPONSIBILITY

- 3 x Program Committee Member**, **3 x Recognized Reviewer Award:** Text2Story @ ECIR '23, '24, '25
- Invited Speaker** on *Advances in NLP* at [AICC 2018](#), Hyderabad.
- Teaching Assistant** for *HSS345: Introduction to Shakespeare* (Spring 2018) and *HSS445: Literature - American Classics* (Spring 2019) at IIIT-Hyderabad.
- Editor-in-Chief** of IIIT-Hyderabad's student-run college magazine, *Ping!* — Aug. 2017 - Aug. 2018
- President** of IIIT-Hyderabad's Literary Club — Aug. 2016 - Aug. 2017
- Guest Reporter** to *The Adivasi Academy, Tejjadgh*, covering Padma Shri Ganesh Devy's [People's Linguistic Survey of India](#), a nationwide initiative documenting 780 languages across India.

WRITING

- Debut poetry collection: [sorting through paperwork after midnight](#), published by Authorspress, May 2019.
- Poetry and prose published in magazines like *Indian Literature* (Issues 275, 286), *Muse India* (Issue 65), and *Madras Courier* (Issues 75, 85). Shortlisted for the [Wingword Poetry Prize, 2021](#).